

Dynamic Grid Trading with Anti-Martingale Sizing and Protective Options: An Extension Toward Bounded Tail Risk

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Note on provenance: This paper extends the dynamic-reset grid trading framework (“DGT”) introduced by Chen, Chen, and Jang [1]. The DGT algorithm and its zero-expectation proof are the original contribution of that work and are cited, not reproduced, below. The extension proposed here — anti-martingale position throttling and an out-of-the-money (OTM) protective put overlay, jointly named DGTAP — together with all backtest results in Sections V–VI, are original to this paper and were produced independently on real market data as described in Section IV.

Abstract—Grid trading strategies profit from price oscillation by placing buy and sell orders on a fixed price lattice. Under a symmetric random-walk assumption, the naive version of this strategy has zero expected profit before fees [1]. Chen, Chen, and Jang [1] proposed Dynamic Grid Trading (DGT), which breaks this zero-expectation result by resetting the grid around the prevailing price whenever it exits the trading band. DGT improves realized return relative to buy-and-hold on Bitcoin and Ethereum, but the same down-break re-capitalization mechanism that gives DGT its edge also concentrates buying activity during sustained downtrends, leaving meaningful residual drawdown risk. We propose two light-touch modifications — (i) a monotonically shrinking (“anti-martingale”) position-sizing multiplier applied to consecutive same-direction grid fills, grounded in the classical fractional-betting literature [11], and (ii) a periodically rolled, deep out-of-the-money protective put funded from a small, fixed fraction of the wallet, grounded in the option-based portfolio insurance literature [12]–[14] — and evaluate the combined strategy (DGTAP) against the DGT baseline on 3.5 years of real minute-resolution Bitcoin/USD data (Bitstamp, January 2021 to July 2024). At an operating point selected through a reset-count-constrained parameter search designed to avoid the overfitting failure modes documented in [24], [25], the anti-martingale throttle alone improves the Calmar ratio from 0.436 to 0.538 with no assumption-dependent inputs. The protective-put overlay’s marginal contribution is shown to be highly sensitive to the assumed relationship between historical realized volatility and market-implied volatility, informed by the empirical crypto variance-risk-premium literature [21]–[23]. We report both an optimistic case (Calmar 1.012) and a conservative case incorporating a typical variance risk premium (Calmar 0.536, effectively breakeven versus throttle-only). We argue the honest contribution of the put overlay cannot be resolved without real options-market data, and report this uncertainty explicitly rather than a single point estimate.

Index Terms—algorithmic trading, grid trading, cryptocurrency, position sizing, anti-martingale, portfolio insurance, options hedging, tail risk, backtest overfitting, Bitcoin.

I. INTRODUCTION

This paper builds on Dynamic Grid Trading (DGT), the strategy proposed by Chen, Chen, and Jang [1] for improving the classical grid trading rule used throughout retail cryptocurrency markets. Cryptocurrency markets have, since the publication of the Bitcoin white paper [2], developed sufficient trading volume and volatility to support a wide ecosystem of retail algorithmic strategies. Grid trading is among the most widely deployed of these: an operator lays down n equally (or geometrically) spaced price levels around a reference price and automatically sells a portion of held inventory as price rises through a level, buying back as it falls through one. Because the strategy is fully mechanical and requires no directional forecast, it is attractive to non-professional traders, and grid-trading modules are built into most major exchanges’ trading-bot interfaces. Related literature on grid-based and market-making systems for foreign exchange and equity markets has explored both hand-crafted rule sets [3] and machine-learning augmented variants that attempt to classify trend regimes before committing capital [4], [5]; more recent work has explored deep sequence models such as LSTM

networks for direct return prediction in financial markets more broadly [6].

Chen, Chen, and Jang [1] formalize the naive version of grid trading with a finite-state absorbing Markov chain and show that, under the assumption that each price step is equally likely to move up or down — consistent with the broader empirical finding that short-horizon asset returns behave close to a martingale under many statistical tests, notwithstanding the well-documented deviations from normality and independence summarized in the “stylized facts” literature [7] — the expected number of completed round trips, and hence the expected profit before fees, is exactly zero (Eq. 3 below, for the symmetric case). Grid trading in its textbook form is therefore not a source of edge; any positive live performance reported by retail users is attributable either to a directional drift in the underlying asset over the sample period, or to the boundary behavior of the grid once price exits the pre-defined band.

DGT [1] directly addresses this boundary case. Instead of halting once price breaks above or below the outer grid lines, DGT re-centers a fresh grid on the current price: a breach above the upper bound triggers full liquidation of held inventory back

to cash, while a breach below the lower bound triggers full deployment of remaining cash into inventory at the new, lower price. This single rule closes the loophole in the zero-expectation proof and, empirically, produces positive risk-adjusted returns on both BTC and ETH over multi-year backtests [1].

We observe, however, that the same mechanism responsible for DGT's edge is also the source of its residual risk. Because each downward grid crossing spends a fixed fraction of remaining cash regardless of how many consecutive downward crossings have already occurred in the current decline, a sustained downtrend causes DGT to behave — locally — like a mild, un-scaled dollar-cost-averaging program. In a genuine bear market or flash crash — a regime cryptocurrency assets are disproportionately prone to, as documented in the Bitcoin volatility literature [8]–[10] — this can leave the strategy heavily invested at a local bottom that is not, in fact, the actual bottom. The published DGT backtest acknowledges this: even after the dynamic-reset fix, maximum drawdown on ETH remains substantial [1].

This paper asks a narrow, practical question: can two standard, independently well-understood risk-management primitives — position-size throttling, of the kind long studied in the capital-allocation and optimal-betting literature originating with Kelly [11], and tail-risk option hedging, of the kind long studied in the portfolio-insurance literature [12]–[14] — be grafted onto DGT with minimal structural change, and if so, at what parameter settings, and with what degree of confidence in the result? We call the resulting strategy DGTAP.

The remainder of this paper is organized as follows. Section II situates DGTAP within four strands of prior literature. Section III restates the DGT mechanism and formally defines the two DGTAP extensions, with twelve numbered equations. Section IV describes the real-market dataset used for evaluation. Section V reports results, including two methodological pitfalls discovered during the parameter search, five tables, and four figures. Section VI discusses limitations, and Section VII concludes.

II. RELATED WORK

A. Rule-Based and Learning-Augmented Grid / Market-Making Systems

Grid and market-making strategies share a common structural idea: liquidity is supplied symmetrically around a reference price, and profit accrues from the level-to-level spread rather than from directional forecasting. Rundo et al. [3] propose a grid trading system robot (GTSbot) for the FX market that augments a conventional grid with a regression-network price forecast and a trend classifier; a companion paper frames the underlying price dynamics as a Markov process to adaptively correct the trading system's parameters over time [4]. Moody and Wu [5] optimize a trading system's parameters

directly against a risk-adjusted objective (the differential Sharpe ratio), conceptually related to our use of the Calmar ratio as a joint objective for the grid, throttle, and put parameters searched in Section V. Aloud and Tsang [15] study agent-based models of FX trading behavior, providing empirical grounding for the kind of short-horizon, rule-following participant a grid strategy implicitly assumes exists on the other side of its trades. Fischer and Krauss [6] show that LSTM networks can extract short-term statistical arbitrage signals from equity markets that decay over time as they become arbitrated away — a cautionary finding for any learned component that might, in future work, be added to DGT's fixed geometric-grid parameters. Avellaneda and Stoikov [16] provide the canonical inventory-risk-aware market-making framework, in which a market maker's quotes are skewed as a function of current inventory; the underlying concern — that a liquidity-supplying strategy accumulates directional inventory risk that a symmetric, static rule set does not price in — is precisely the concern motivating the anti-martingale throttle of Section III-B.

B. Option-Based and Dynamic Portfolio Insurance

The idea of bounding a portfolio's downside by combining a risky position with a protective derivative dates to the option-based portfolio insurance (OBPI) framework of Leland and Rubinstein [12]. Black and Jones [13] and Black and Perold [14] subsequently proposed constant proportion portfolio insurance (CPPI) as a derivative-free alternative, dynamically rebalancing between a risky “cushion” and a risk-free floor asset. CPPI avoids explicit option-premium cost at the expense of requiring continuous rebalancing and being more exposed to discontinuous (jump) price moves between rebalancing points, a concern formalized in the jump-diffusion option-pricing extension of Merton [17]. DGTAP's put overlay (Section III-C) is structurally an OBPI-style approach, chosen over a CPPI-style scheme specifically because DGT's own down-break mechanism already performs a CPPI-like full reallocation at the lower grid boundary, and it is precisely the gap risk between boundary-breach events that a rebalancing-only scheme cannot protect against but an already-purchased option can. The Black–Scholes pricing framework used to value the put overlay is due to Black and Scholes [18].

C. Cryptocurrency Volatility and the Pricing of Crypto Options

Because the put overlay's cost is a direct function of assumed volatility, its evaluation depends on the literature characterizing realized and implied volatility in crypto markets. The ARCH framework of Engle [19], generalized by Bollerslev [20], underlies most subsequent volatility-clustering studies, including several applied to Bitcoin: Katsiampa [8] compares GARCH-family specifications for Bitcoin return volatility and finds strong volatility clustering and asymmetry, while Dyhrberg [9], [10] documents Bitcoin's volatility and hedging

properties relative to gold and the US dollar. More directly relevant to the put overlay's cost, Almeida et al. [21] estimate risk premia in the Bitcoin options market and find a substantially higher variance risk premium than in the S&P 500, consistent with our use of a $\lambda > 1$ multiplier in the conservative scenario of Sections V-D–V-E. Du et al. [22] propose a Bitcoin option pricing model incorporating volatility-of-volatility dynamics and find the crypto variance risk premium can vary in sign over time — a finding that argues against treating λ as a single fixed constant, a simplification flagged in Section VI. Cao and Celik [23] separately study the valuation of traded Bitcoin options, providing further empirical grounding for the gap between naive historical-volatility pricing and observed market premia.

D. Backtest Overfitting and Risk-Adjusted Performance Measurement

A parameter search of the kind performed in Section V is only as credible as the safeguards taken against reporting a result that is an artifact of having tried many configurations against the same historical path. Bailey and López de Prado [24] formalize this concern with the Deflated Sharpe Ratio; Bailey, Borwein, López de Prado, and Zhu [25] propose a combinatorially symmetric cross-validation procedure to directly estimate the probability that an apparently best-performing configuration would underperform out-of-sample. We do not apply either technique formally, but adopt their qualitative warning directly: Section V-B documents a case in which unconstrained optimization degenerated toward configurations supported by too few independent events to be statistically meaningful, and we impose a minimum-event-count constraint as a partial safeguard. For summarizing risk-adjusted performance, we use the Calmar ratio (Eq. 11); the relationship between maximum drawdown, the Sharpe ratio [26] (Eq. 12), and sample length is characterized by Magdon-Ismail and Atiya [27] and, in continuous time, by Magdon-Ismail, Atiya, Pratap, and Abu-Mostafa [28] — a further reason for the minimum-event-count filter applied in Section V-B.

III. DGTAP: PROPOSED EXTENSIONS

A. Review of the DGT Mechanism

We restate the DGT mechanism at the level needed to define our extension; see [1] for the full absorbing-Markov-chain derivation. A geometric grid is defined by a center price P_0 , a per-level ratio k , and a half-width m , producing $2m+1$ levels:

$$P_i = P_0(1+k)^i, \quad i \in \{-m, \dots, m\} \quad (1)$$

As price rises through level i (counted toward the upper boundary), the strategy sells a fraction of currently held inventory Q_{held} ; as price falls through a level j levels above the lower boundary, it buys using a fraction of remaining cash W_{cash} , where G_i and G_j denote the number of levels remaining

to the respective boundary:

$$\text{Sell}_i = \frac{1}{G_i} Q_{\text{held}}, \quad \text{Buy}_j = \frac{1}{G_j} W_{\text{cash}} \quad (2)$$

This harmonic-fraction rule is what gives the naive grid its zero-expectation property under [1]'s Markov-chain analysis, formally:

$$\mathbb{E}[N|n, m] = \frac{n^2}{4} \quad (\text{naive grid, symmetric random walk}) \quad (3)$$

where N is the number of completed round trips and $n = 2m$ is the total level count. DGT's contribution is the boundary rule: on breaching $P_0(1+k)^m$ (upper) or $P_0(1+k)^{-m}$ (lower), the strategy does not stop. It fully converts to cash (upper breach) or fully converts to inventory (lower breach), then re-instantiates a fresh grid centered on the breach price. This is repeated for the life of the backtest.

B. Anti-Martingale Position Throttle

We define a consecutive-fill counter c that increments by one on every down-crossing buy and resets to zero on any up-crossing sell or grid reset. The buy fraction actually executed at the c -th consecutive down-crossing is:

$$f_c = \frac{1}{G_j} \cdot \mu(c) \quad (4)$$

$$\mu(c) = \max(\text{floor}, \text{start} - \text{step} \cdot (c - 1)) \quad (5)$$

with $\text{start}, \text{floor} \in (0, 1]$ and $\text{floor} < \text{start}$. The multiplier μ starts at start on the first fill of a new down-run and shrinks linearly, by step per additional consecutive fill, until it reaches floor , where it remains. We term this “anti-martingale” in the specific and limited sense that it is a monotonic de-risking of grid trading's inherent buy-the-dip repetition, not a literal reversal of the martingale betting system studied in the classical fractional-betting literature [11] — there is no doubling; exposure per event can only shrink, never grow. This is deliberately the opposite prescription from a Kelly-style scheme in a genuinely favorable-odds repeated bet [11], where the growth-optimal policy holds the wagered fraction constant or increases it with estimated edge; here each additional consecutive down-fill is treated as evidence against, not in favor of, continuing to commit capital at the un-throttled rate. Figure 2 plots $\mu(c)$ for three candidate schedules, including the Calmar-optimal schedule identified in Section V-C.

C. OTM Protective Put Overlay

At every grid reset event (and optionally on a fixed calendar cadence), the strategy allocates a fixed fraction b of current wallet cash to purchase a European put struck $\text{otm}\%$ below spot, with tenor τ , in the spirit of option-based portfolio insurance [12]. The premium is estimated via the Black–Scholes formula [18]:

$$\text{Put}(S, K, \tau, \sigma) = Ke^{-r\tau}N(-d_2) - SN(-d_1) \quad (6)$$

with:

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau} \quad (7)$$

using a trailing realized-volatility estimate over lookback window L and bar frequency B , optionally scaled by an assumed implied-to-realized volatility ratio $\lambda \geq 1$ to approximate the variance risk premium documented empirically in the Bitcoin options market [21]–[23]:

$$\sigma = \lambda \hat{\sigma}, \quad \hat{\sigma} = \text{std}(r_{t-L:t}) \cdot \sqrt{B} \quad (8)$$

We do not attempt to model volatility skew or the volatility-of-volatility dynamics found by [22] to affect the sign of the crypto variance risk premium over time; a flat- λ approximation is a simplification revisited in Section VI. At expiry, the position pays $\max(K - ST, 0)$ per unit notional, credited back to the wallet; a new put is then rolled at the next qualifying event. Because the position is deep out-of-the-money and small relative to total capital, its purpose is explicitly tail insurance rather than income generation, addressing the gap-risk scenario that DGT's own down-break rule cannot cover between discrete reset events, and which Merton's jump-diffusion analysis [17] shows a continuously-rebalanced (CPPI-style) scheme is structurally unable to fully hedge.

D. Combined Algorithm and Evaluation Metrics

DGTAP is DGT with the buy-side fraction replaced by f_c from Eq. 4 and an independent put overlay (Section III-C) running in parallel, funded out of the same wallet. The two mechanisms are structurally independent — the throttle never touches the option book and the put overlay never touches grid sizing — so their marginal contributions are evaluated both separately and jointly in Section V. Throughout, performance is summarized with annualized IRR, maximum drawdown (MDD), and the Calmar ratio:

$$\text{IRR} = \left(\frac{V_T}{V_0}\right)^{1/T_{yr}} - 1 \quad (9)$$

$$\text{MDD} = \min_t \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \quad (10)$$

$$\text{Calmar} = \frac{\text{IRR}}{|\text{MDD}|} \quad (11)$$

where V_0 and V_T are initial and final portfolio value, T_{yr} is the sample length in years, and V_t is the mark-to-market equity curve. For reference, the classical Sharpe ratio [26] used throughout the risk-adjusted performance literature reviewed in Section II-D is:

$$\text{Sharpe} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p} \quad (12)$$

TABLE I

Summary of Symbols Used in Section III

Symbol	Meaning
P_0, k, m	Grid center price, level ratio, half-width
G_i, G_j	Levels remaining to upper / lower boundary
$c, \mu(c)$	Consecutive down-fill count; throttle multiplier
start, floor, step	Throttle schedule parameters
S, K, τ, σ	Spot, strike, tenor, volatility (put pricing)
λ	Assumed implied / realized volatility ratio
b, otm	Put premium budget fraction; OTM depth
$V_t, \text{IRR}, \text{MDD}$	Equity curve; annualized return; max drawdown

IV. DATA AND EXPERIMENTAL SETUP

All results in this paper use real, exchange-sourced data: Bitstamp BTC/USD, one-minute resolution, January 1, 2021 through July 31, 2024 (1,882,081 bars), matching the evaluation window of [1]. This is a deliberate departure from purely synthetic testing: geometric Brownian motion paths priced with Black–Scholes [18] and evaluated with the same volatility model produce circular, overly favorable put-overlay results, since GBM contains no variance risk premium of the kind documented empirically for Bitcoin options [21]–[23] and often no jump risk not already reflected in trailing realized volatility, contra the jump-diffusion dynamics analyzed by Merton [17] and the volatility-clustering, fat-tailed “stylized facts” documented broadly by Cont [7]. Working from one real historical path removes that circularity but introduces the opposite concern — a single asset over a single multi-year window is a small sample, central to the backtest-overfitting literature reviewed in Section II-D [24], [25]. We treat this limitation explicitly in Section VI.

A round-trip taker fee of 8 bps (0.0008) is applied to every grid fill, consistent with typical spot exchange fee tiers. Initial capital is normalized to 100 units for all reported figures.

V. RESULTS

A. Pitfall 1: Resampling Bias in Grid-Parameter Search

An initial parameter search was conducted on hourly-resampled data to reduce computation time. This produced materially different — and, on validation, worse — optimal grid parameters than a search conducted directly on native one-minute data. Grid trading's edge is generated by the count of price crossings through fine-grained levels, a quantity that hourly closes systematically undercount. Coarse proxies are unsafe specifically for the grid-size and level-count dimensions of the parameter space, even though they remain informative for dimensions unrelated to intra-bar path dependence (e.g., the relative ranking of option strike/tenor choices, Section V-D). All grid-size results below were obtained on native one-minute data.

B. Pitfall 2: Unconstrained Optimization Degenerates to Zero-Trade Regimes

An unconstrained sweep over grid size and half-width, using Calmar ratio as the objective, monotonically favored wider grids, with the nominal optimum occurring at widths so large the grid never reset over the entire 3.5-year sample. Such configurations are not meaningfully testing the DGT mechanism; they degenerate to an almost-static spot position whose apparent performance is driven by a handful of large, sample-specific price moves — exactly the small-effective-sample-size failure mode the backtest-overfitting and drawdown-scaling literature warns against [24], [25], [27], [28]. We therefore constrained the search to configurations with at least 40 grid-reset events over the sample window.

Fig. 1. Calmar ratio, grid $\times$ half-width (throttle active, 1-min BTC data)

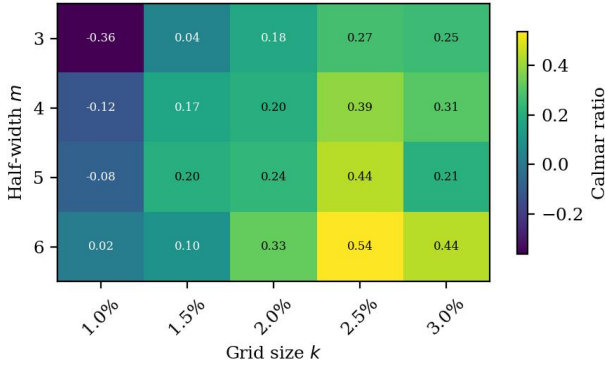


Fig. 1. Calmar ratio as a function of grid size k and half-width m , with the anti-martingale throttle active, evaluated on native 1-minute BTC data under the 40-reset constraint of Section V-B.

C. Anti-Martingale Throttle: Parameter Search and Result

Within the reset-constrained space, a grid search over (start, floor, step) identified start = 0.8, floor = 0.4, step = 0.15 as Calmar-optimal (Fig. 2). This result does not depend on any option-pricing assumption and is treated as comparatively high-confidence.

Fig. 2. Anti-martingale throttle schedules

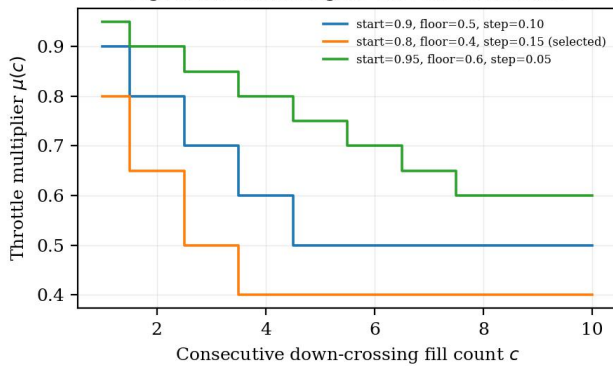


Fig. 2. Throttle multiplier $\mu(c)$ for three candidate schedules; the selected schedule (orange) reaches its floor fastest, most aggressively de-risking

sustained down-runs.

TABLE II

Throttle Isolated From Put Overlay (BTC, 1-min, 2021-01–2024-07)

Strategy	IRR	MDD	Calmar	Resets
Buy & Hold	25.93%	-77.54%	0.334	—
DGT baseline ($k=2.5\%$, $m=6$)	20.20%	-46.29%	0.436	73
DGT + throttle	21.09%	-39.17%	0.538	73

The throttle improves both legs of the Calmar ratio simultaneously: return is essentially unchanged (+0.89 pp) while drawdown improves by 7.12 percentage points. Because consecutive-fill throttling only reduces exposure taken during an already-adverse run, and the improvement holds at fixed grid parameters and trade timing, we consider this the most robust finding of the paper.

D. OTM Put Overlay: Relative Parameter Efficiency

Because the put overlay's absolute profitability is a direct function of an unobservable assumption (λ), we depart from absolute Calmar optimization — which degenerates trivially to a zero budget under any realistic λ — and instead search for the strike/tenor combination that is relatively most capital-efficient, holding premium budget fixed, and that ranks consistently well under both $\lambda = 1.0$ and $\lambda = 1.4$ (Fig. 3).

Fig. 3. Net hedge P&L (% of principal) by strike/tenor

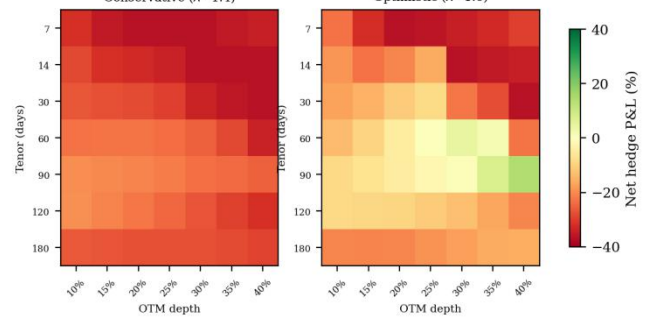


Fig. 3. Net hedge P&L (% of principal) across OTM depth and tenor, conservative (left) vs. optimistic (right) volatility assumptions. The 90-day tenor column is consistently the least-negative / most-positive band in both panels.

A joint sweep initially produced implausible drawdown figures (MDD approaching -100%) for short-tenor, cheap-premium configurations, traced to a leverage artifact: very cheap premiums purchase very large option notional for a fixed dollar budget, and that notional's mark-to-market value swings disproportionately with spot. We report this as a limitation of Black-Scholes-based [18] backtest marking under a fixed-dollar-budget rule, not a property of real options markets. Results below use a reduced, non-degenerate budget regime.

TABLE III

Top-10 Intersection by Net Hedge P&L Across $\lambda = 1.0$ and $\lambda = 1.4$

Tenor	Robust in both scenarios?	Notes
90 days	Yes	Top-10 across 15–30% OTM in both rankings
7–14 days	No	Excess roll cost; marking artifact risk
120–180 days	No	Premium staleness; larger vega exposure

Strike depth is where the two assumptions diverge: the conservative case favors closer-to-the-money strikes (15–20% OTM), consistent with the elevated variance-risk-premium estimates in [21]; the optimistic case favors deeper OTM strikes (35–40%), which capture large convex payoffs during the sample's sharpest declines but whose value is most sensitive to the pricing assumption. 20–25% OTM at 90-day tenor is proposed as the parameter choice least likely to be regretted under either assumption, though optimal under neither.

E. Combined DGTAP Operating Point

TABLE IV

Grid 2.5%/m=6, Throttle 0.8/0.4/0.15, Put 20% OTM/90-day

Strategy	IRR	MDD	Calmar
Buy & Hold	25.93%	−77.54%	0.334
DGT (baseline of [1])	20.20%	−46.29%	0.436
DGT + throttle	21.09%	−39.17%	0.538
DGTAP, conservative ($\lambda=1.4$)	20.48%	−38.20%	0.536
DGTAP, optimistic ($\lambda=1.0$)	21.92%	−21.65%	1.012

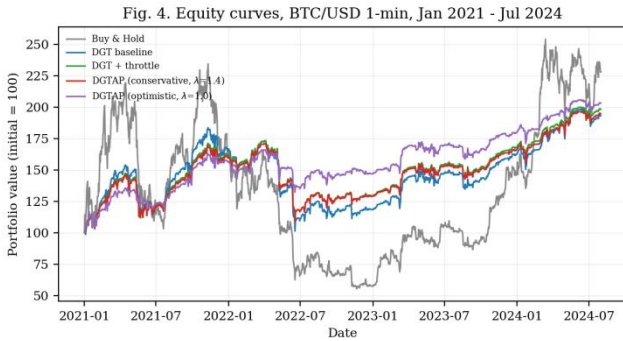


Fig. 4. Mark-to-market equity curves for all five strategies of Table IV, daily-resampled, BTC/USD, Jan 2021–Jul 2024. DGTAP (optimistic) visibly decouples from the other DGT-family curves after mid-2022; DGTAP (conservative) tracks throttle-only closely.

Under the conservative assumption, the put overlay is close to a wash relative to throttle-only (Calmar 0.536 vs. 0.538). Under the optimistic assumption, it nearly doubles the Calmar ratio. We do not average or otherwise collapse these two figures into a single point estimate, because doing so would misrepresent a genuine and currently irreducible modeling uncertainty as statistical noise — precisely what the deflated-Sharpe and backtest-overfitting literature [24], [25] cautions against.

TABLE V

Ranges Tested in Section V

Parameter	Range tested	Selected
Grid size k	1.0% – 15%	2.5%
Half-width m	3 – 8	6
Throttle start	0.8 – 0.95	0.8
Throttle floor	0.4 – 0.6	0.4
Throttle step	0.05 – 0.15	0.15
OTM depth	10% – 40%	20–25%
Tenor τ	7 – 180 days	90 days
λ (IV/RV ratio)	1.0 – 1.4	reported as range

VI. DISCUSSION AND LIMITATIONS

Single-asset, single-window evaluation: All results use BTC/USD only, over one historical period. The original DGT study [1] additionally evaluates ETH; we were unable to obtain comparable real minute-resolution ETH data within the constraints of this study and consider ETH validation, and validation across multiple non-overlapping historical windows, a prerequisite before the reported parameter values should be treated as settled recommendations. Bitcoin and Ethereum exhibit differing volatility-clustering and asymmetry dynamics even within the same GARCH-family framework [8], so parameter values tuned on BTC should not be assumed to transfer.

Option pricing is simulated, not observed: The put overlay is priced with Black–Scholes [18] from trailing realized volatility, not an actual options order book. Real crypto OTM puts also exhibit volatility skew and a time-varying, sometimes sign-switching variance risk premium [21]–[23] that a flat- λ model does not capture. We regard the 90-day, 20–25% OTM operating point as a structural hypothesis to be tested against real quoted implied volatility surfaces, not a validated conclusion.

Overfitting risk in grid parameter selection: Section V-B documents that naive optimization of grid width degenerates toward configurations with too few reset events to be statistically meaningful; the 40-reset floor is a heuristic guard, considerably less rigorous than the formal frameworks of [24], [25]. Reporting a single “best” configuration (Table IV) necessarily understates the uncertainty around that choice; a wider ensemble of near-optimal configurations, evaluated out-of-sample and scored with the length-normalized corrections of [27], [28], would be more defensible.

Throttle result is the most defensible finding of this paper: it requires no external pricing model and shows a consistent, simultaneous improvement in both return and drawdown across the parameter neighborhoods tested (Calmar 0.436 $\rightarrow$ 0.538). The put overlay's contribution should be read as a bounded range (roughly breakeven to a near-doubling of Calmar) pending real market data.

Relationship to related algorithmic-trading paradigms: The anti-martingale throttle and the inventory-skewing quotes of market-making frameworks such as Avellaneda–Stoikov [16] address a related concern — a liquidity-supplying strategy's tendency to accumulate directional inventory — through different mechanisms: theirs continuously adjusts quote placement via a formal risk-aversion parameter, while ours applies a simpler, discrete, rule-based multiplier tied to DGT's existing fixed-fraction buy rule. A natural future direction, informed by the demonstrated (if decaying) predictive power of sequence models in related financial prediction tasks [6], would replace the throttle's fixed linear decay with a learned function of recent realized volatility or order-flow imbalance; we did not pursue this here in order to keep the throttle's contribution free of additional model-dependent assumptions.

VII. CONCLUSION

Building directly on the DGT mechanism of [1], we proposed two structurally independent extensions targeting DGT's residual drawdown risk: a consecutive-fill anti-martingale position throttle grounded in the classical fractional-betting literature [11], and a periodically rolled OTM protective put grounded in the option-based portfolio insurance literature [12]–[14], [17], [18]. On 3.5 years of real Bitcoin minute-resolution data, the throttle produces a robust, assumption-free improvement in risk-adjusted return (Calmar ratio +0.10 at the identified operating point). The put overlay's marginal value is real but genuinely uncertain, ranging from roughly neutral to strongly positive depending on an option-pricing assumption this study cannot resolve without access to actual options-market quotes — an uncertainty the empirical crypto variance-risk-premium literature [21]–[23] suggests is unlikely to be resolved by a single fixed multiplier in any case. We regard the honest reporting of that uncertainty, together with the two methodological pitfalls documented in Sections V-A–V-B, as being as valuable a contribution of this paper as the point estimates themselves. Future work should validate the proposed operating point against real options-implied volatility surfaces, extend the evaluation to ETH and to multiple non-overlapping historical windows, and formally test the statistical significance of the throttle's drawdown improvement against a resampled/bootstrapped null using the methodology of [24] and [25].

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APPENDIX A CODE AND REPRODUCIBILITY

The DGTAP backtesting implementation (grid engine, anti-martingale throttle, Black–Scholes put overlay [18], and all parameter sweeps referenced in Section V) is implemented in Python (`dgtap_backtest.py`) and was run against the Bitstamp dataset cited in Section IV. The implementation reproduces the DGT reset mechanism from first principles based on the published algorithm description in [1] rather than reusing the released source code accompanying that paper, and may therefore differ in minor implementation details. Absolute performance figures for the DGT baseline reported here should not be assumed to exactly match the figures originally reported in [1] for this reason; relative comparisons between DGT and DGTAP within this paper, run on identical code paths and identical data, are the results this paper's conclusions rely on.